MA 301: Test 2 Practice

Some Formulas
* Let $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$ - $z_1 z_2 = r_1 r_2 \left[\cos(\theta_1+\theta_2) + i\sin(\theta_1+\theta_2)\right]$ - $\dfrac{z_1}{z_2} = \dfrac{r_1}{r_2}\left[\cos(\theta_1-\theta_2) + i\sin(\theta_1-\theta_2)\right] \quad (z \neq 0)$ * Let $z = r(\cos\theta + i\sin\theta)$ - $z^n = r^n\left[\cos(n\theta) + i\sin(n\theta)\right]$ - $z^{1/n} = r^{1/n}\left[\cos\left(\dfrac{\theta+2\pi k}{n}\right) + i\sin\left(\dfrac{\theta+2\pi k}{n}\right)\right]$ for $k = 0, 1, 2, \ldots, n-1$

Problem 1: Let C be the boundary of the disc $y^2 + z^2 \le 9$ in the $x = 2$ plane.

Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$ for the vector fields: (a) $\mathbf{F} = [z^3, x^3, 3y]$ and (b) $\mathbf{F} = [z^3, x^3, y^2]$

Problem 2: Let $F = [y^2, x^2, z + x]$ and C be the boundary of the triangle with vertices $(0, 0, 0), (1, 0, 0), (1, 1, 0)$ Compute $\oint_C \mathbf{F} \cdot d\mathbf{r}$

Problem 3: Let $F = [0, z^3, 0]$ and C be the boundary of the cylinder $x^2 + y^2 = 1,\ x \geq 0,\ y \geq 0,\ 0 \leq z \leq 1$
Compute $\oint\limits_C \mathbf{F} \cdot d\mathbf{r}$

Problem 4: Let $z_1 = 5 + i$, $z_2 = 2 - 4i$, $z_3 = a + bi$.
Perform the following operations and write the result in standard form.

(a) $z_1 + z_2$

(b) $z_1 z_3$

(c) $\dfrac{z_2}{z_1}$

(d) $|z_2|$

(e) $z_1 - iz_2$

(f) $\sqrt{-10} \cdot \sqrt{-6}$

Problem 5: Let $z_1 = 1 - i$, $z_2 = 3i$. Answer the following:

(a) Find the polar form of z_1 and plot it.

(b) Find the polar form of z_2 and plot it.

(c) Find z_1^{12}.

(d) Find $z_1^{1/3}$.

(d) Solve for z: $z^4 + z_1 = z_2$